

# CM 7028 Digital Media Production

Chutisant Kerdvibulvech, Ph.D.  
Assistant Professor

ผศ.ดร.ชุตินันต์ เกิดวิบูลย์เวช

# CHATBOT

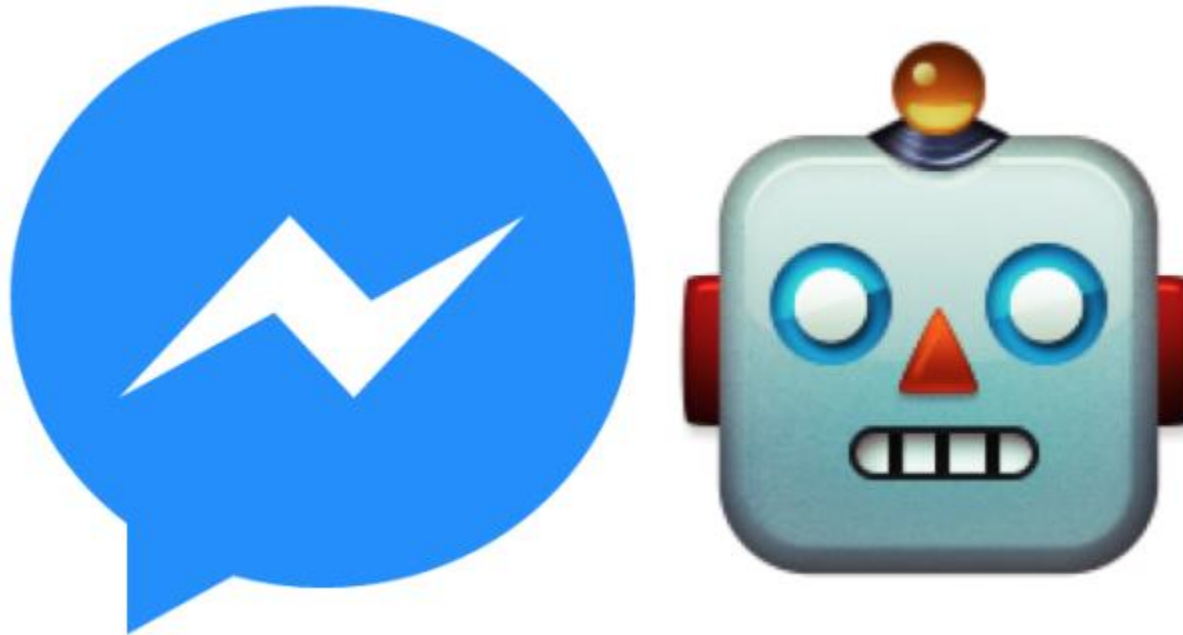


<http://knowledge.wharton.upenn.edu/article/rise-chatbots-time-embrace/>

“A chatbot (also known as a talkbot, chatterbot, Bot, chatterbox, Artificial Conversational Entity) is a computer program which conducts a conversation via auditory or textual methods.”

<https://en.wikipedia.org/wiki/Chatbot>

# CHATBOT = CHAT + ROBOT



<http://www.garethjames.net/complete-guide-chatbots/>

# CHATBOT: HOW DOES IT WORK?

5

#virtual assistant #virtual agent #bot

## How does it work?



User makes a request



Request analyzed  
by artificial intelligence  
User info taken into account  
(history, preferences...)



Real time response  
Conversational strategy  
Interactions personalization

<http://www.garethjames.net/complete-guide-chatbots/>

# MACHINE LEARNING

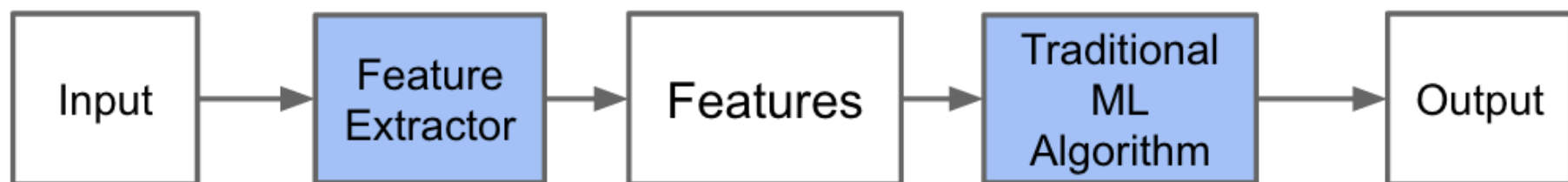
- ⦿ “Machine learning is the subfield of computer science that gives computers the ability to learn without being explicitly programmed” (Arthur Samuel, 1959).
- ⦿ “Evolved from the study of pattern recognition and computational learning theory in artificial intelligence,[2] machine learning explores the study and construction of algorithms that can learn from and make predictions on data” Ron Kohavi; Foster Provost (1998). "Glossary of terms". Machine Learning. 30: 271-274.

# DEEP LEARNING

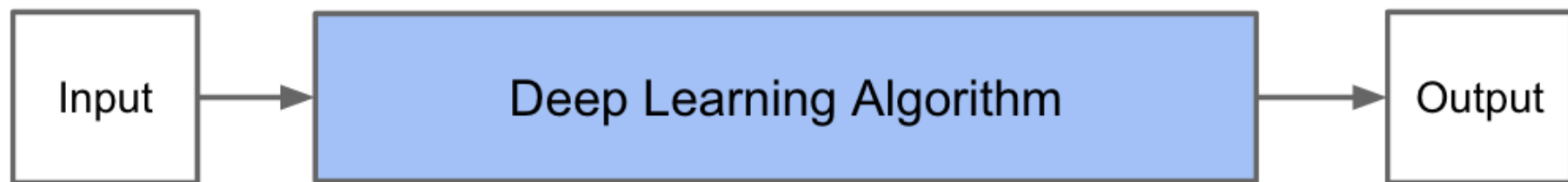
- ⦿ “Deep learning (also known as deep structured learning, hierarchical learning or deep machine learning) is a branch of machine learning based on a set of algorithms that attempt to model high level abstractions in data.”
- ⦿ “Various deep learning architectures such as deep neural networks have been applied to fields like computer vision, automatic speech recognition, and natural language processing”

[https://en.wikipedia.org/wiki/Deep\\_learning](https://en.wikipedia.org/wiki/Deep_learning)

# MACHINE LEARNING VS DEEP LEARNING



Traditional Machine Learning Flow

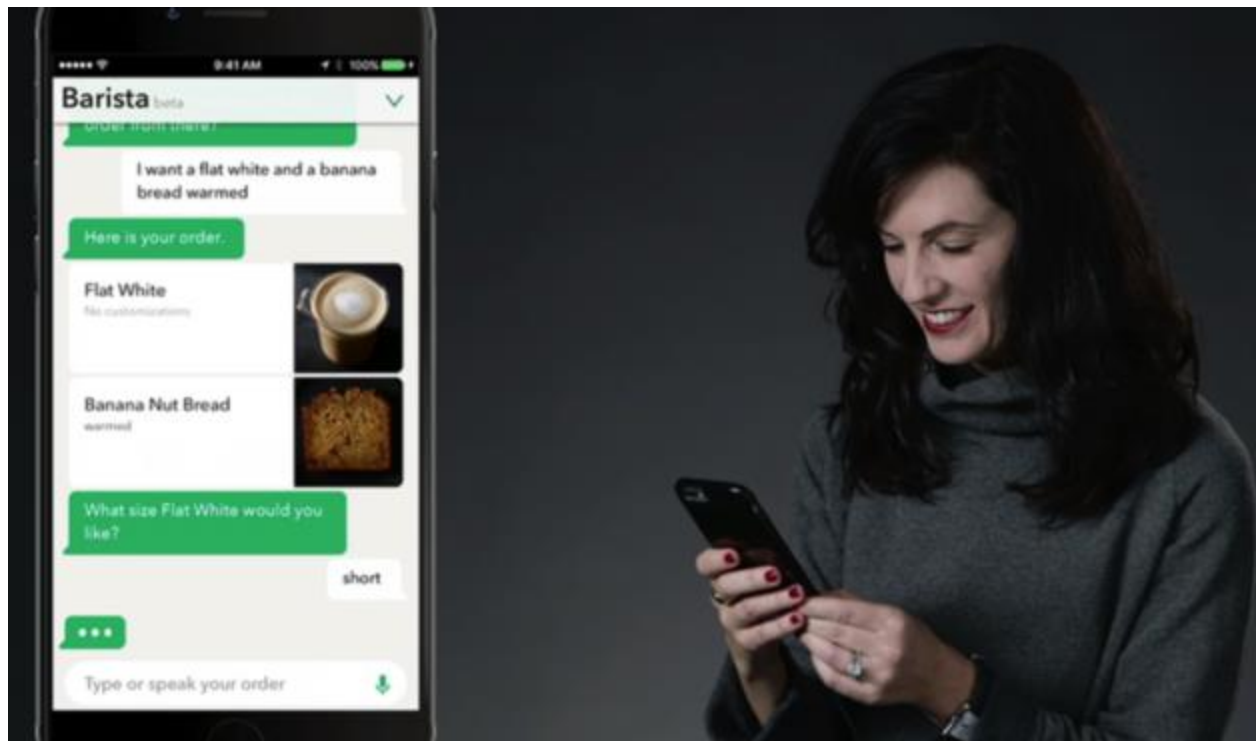


Deep Learning Flow

<http://adilmoujahid.com/posts/2016/06/introduction-deep-learning-python-caffe/>

# STARBUCKS UNVEILS CHATBOT APP (1)

- ◉ A voice-activated chatbot, called MyStarbucks Barista



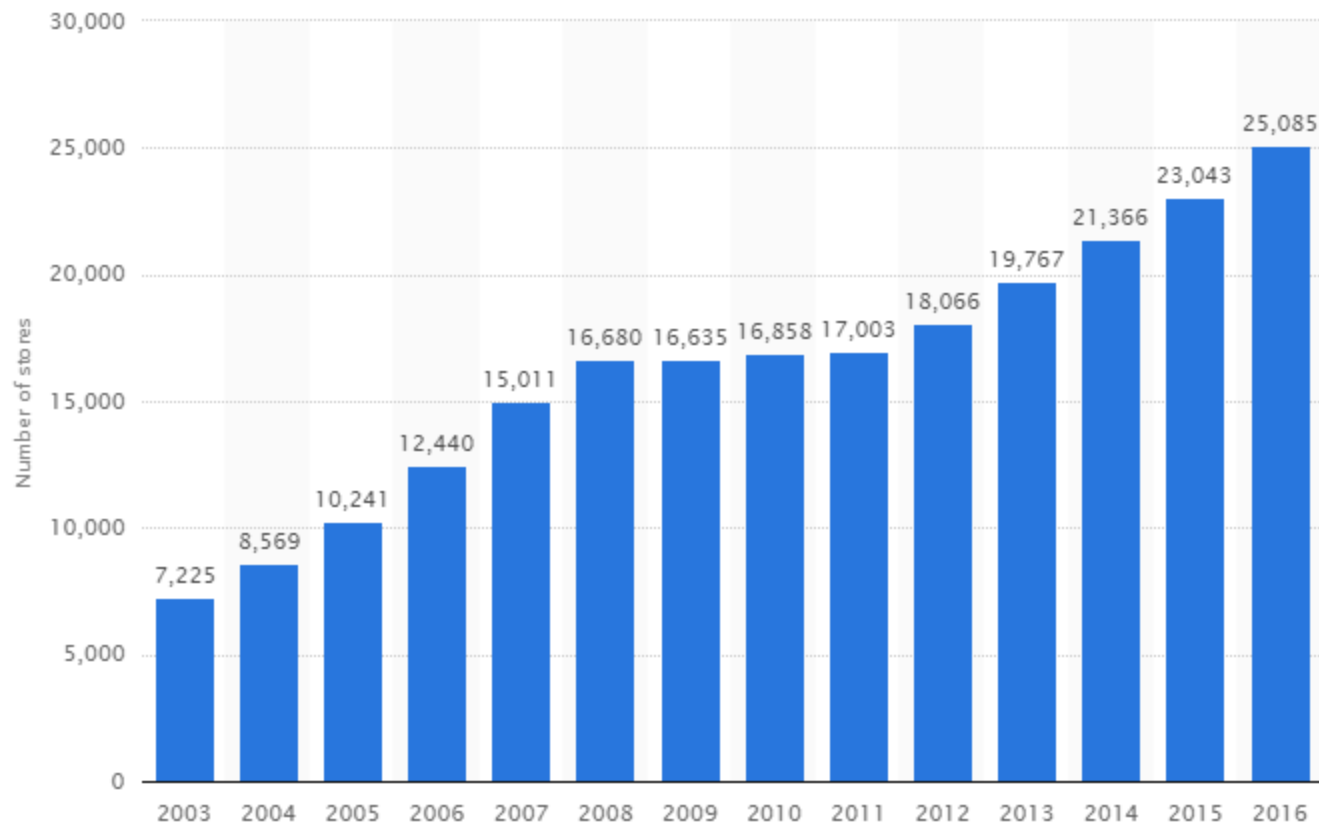


# STARBUCKS UNVEILS CHATBOT APP (2)

- ⦿ Let customers place orders by speaking, "I want a latte." The features can also be used to pay for orders.
- ⦿ MyStarbucks Barista is only available to 1,000 people before a nationwide release this summer.

# STARBUCKS UNVEILS CHATBOT APP (3)

Number of Starbucks stores worldwide from 2003 to 2016



<https://www.statista.com/statistics/266465/number-of-starbucks-stores-worldwide/>

Chutisant Kerdvibulvech

# MELODY (1)

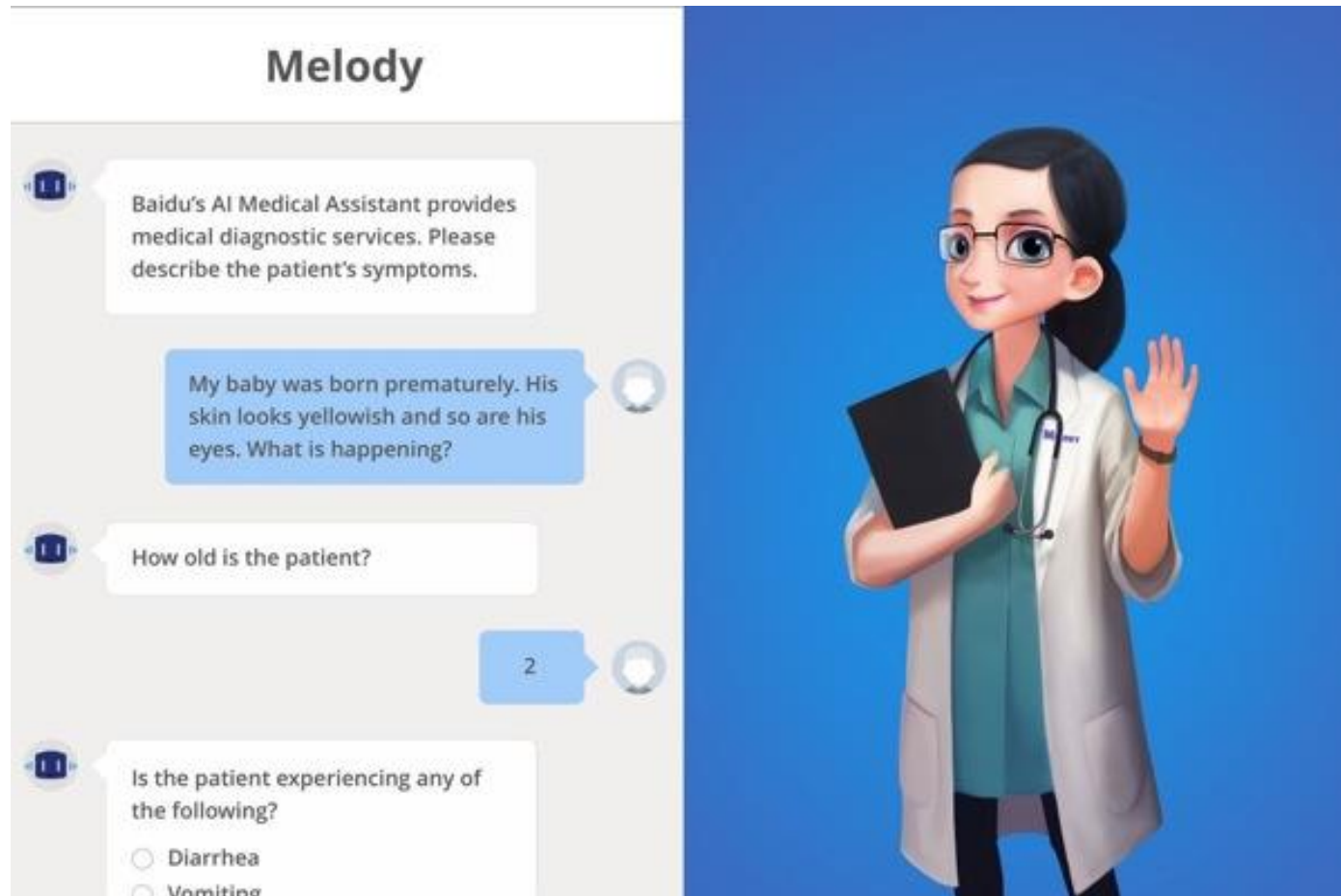


<http://www.cnbc.com/2016/10/11/chinese-baidu-unveils-ai-health-chatbot-for-patients-and-doctors.html>

- ◉ Baidu, China's biggest search engine
- ◉ Launches medical chatbot to help Chinese doctors diagnose patients

## MELODY (2)

- Built into the company's iOS and Android Baidu Doctor app, which launched in China in 2015



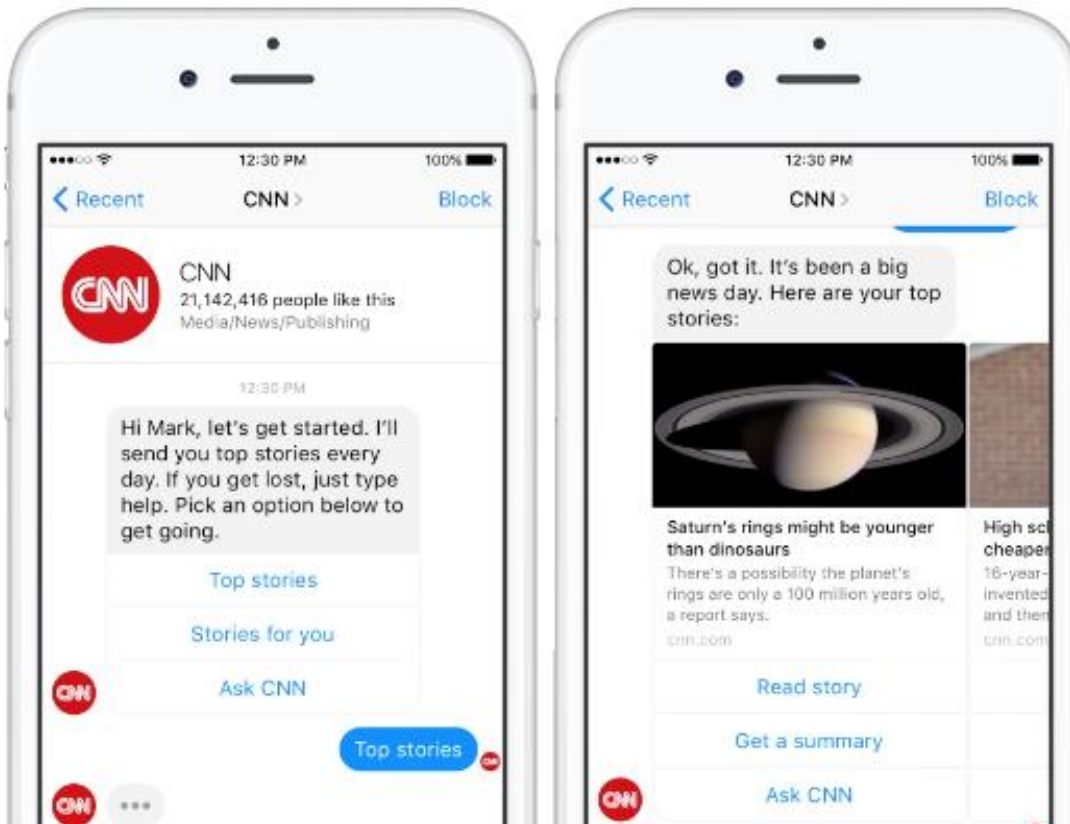
<http://venturebeat.com/2016/10/11/baidu-launches-medical-assistant-chatbot-to-help-doctors-collect-patient-information/>

# BOTS FOR MESSENGER: CNN (1)

- ⦿ “Facebook announced at its F8 conference that it has created a platform for companies to develop the robots that mimic human conversation to run within its Messenger app”
- ⦿ “Bots for Messenger enables users to get everything from weather updates to shopping notifications and personalized news as well as message companies and publishers as they do their friends.”

<http://www.brandchannel.com/2016/04/14/brand-chatbots-041416/>

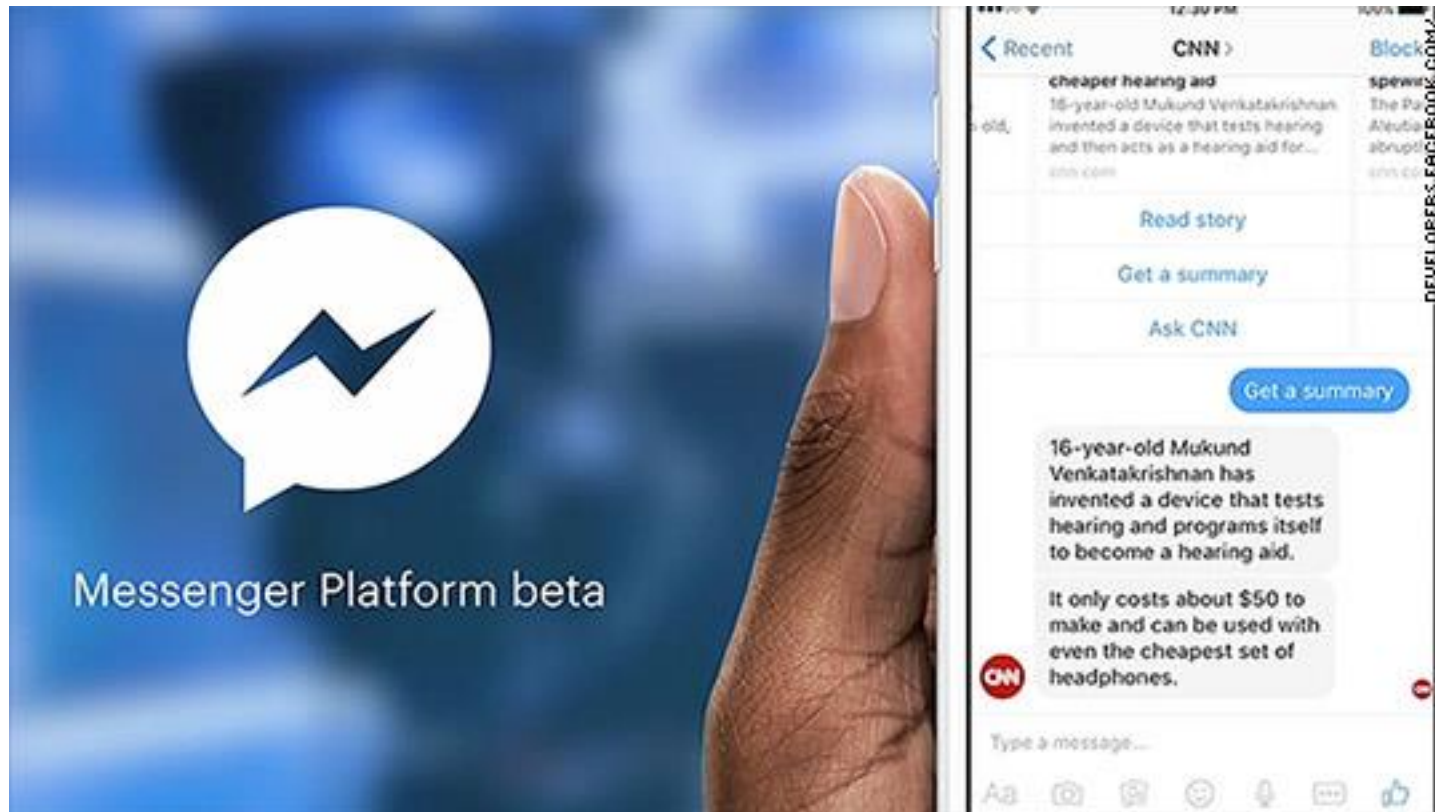
# BOTS FOR MESSENGER: CNN (2)



- “Launch partners include shopping app Spring, HP and CNN.”

<http://www.brandchannel.com/2016/04/14/brand-chatbots-041416/>

# BOTS FOR MESSENGER: CNN (3)



<http://money.cnn.com/2016/04/12/technology/facebook-messenger-bots/>

# INTEGRATED GROUP PROJECT

- **Virtual Reality**
- **Facebook 360**
- **LINE Application**
- **Chatbot**



# QUESTIONS?

Asst. Prof. Dr. Chutisant Kerdvibulvech

Graduate School of Communication Arts and Management Innovation,  
National Institute of Development Administration (NIDA)

[chutisant.ker@nida.ac.th](mailto:chutisant.ker@nida.ac.th)

<https://www.facebook.com/chutisant.kerdvibulvech>